

Practical #2

Practical Title:

To understand version control concepts and perform basic source code management using Git

Theory:

1. What is Version Control System (VCS)?

A Version Control System (VCS) is a tool that helps track changes in source code over time. It allows multiple developers to work together and manage different versions of a project.

2. Types of Version Control Systems

◆ Centralized VCS

- Single central server stores code
- Example: SVN
- Requires internet to access

◆ Distributed VCS

- Each user has a complete copy of repository
 - Example: Git
 - Works offline and faster
-

3. What is Git?

Git is a **distributed version control system** used to track code changes, collaborate with teams, and manage project history efficiently.

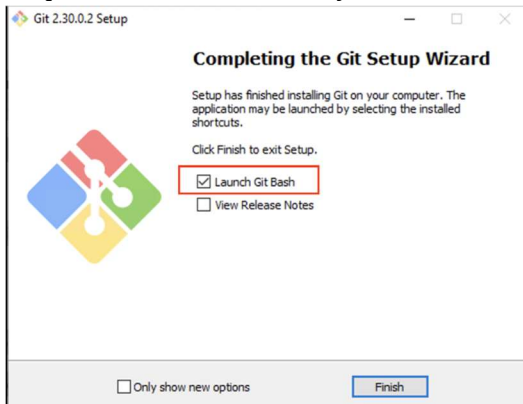
4. Key Git Terminologies

- **Repository:** Storage location of project files and history
- **Commit:** Saving changes with a message

- **Branch:** Separate line of development
- **Merge:** Combining branches
- **Clone:** Copying a repository to local system

Procedure:

Step 1: Install Git and verify installation



```
git --version

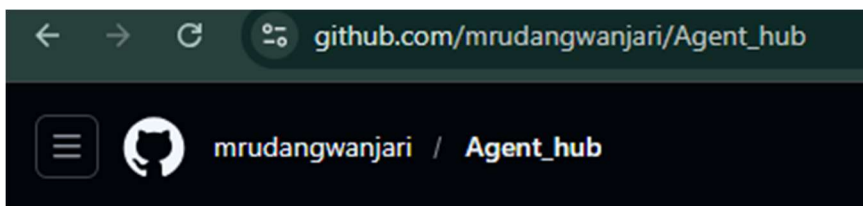
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Step 2: Configure Git

```
$ git init
Reinitialized existing Git repository in C:/Users/MRUDANG WANJARI/Agent_Hub/.git/
```

Step 3: Create a Repository



Step 4: Create a file and add content

```
<html>
  <body>
    <h1>Hello </h1>
  </body>
</html>
```

Step 5: Add and Commit

```
$ git add .
$ git commit -m "first commit"
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
```

Step 6: Connect to GitHub and Push

```
$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
```

Step 7: Pull changes

```
$ git pull
Already up to date.
```

```
$ git fetch
From https://github.com/mrudangwanjari/Agent_hub
* [new branch]      main      -> origin/main
```

Activity / Task:

Task 1:

Create your own GitHub repository and push at least one file.

Repository Link:

<https://github.com/sharvayuzade/DevOps-SEM-VI>

Task 2:

Create a new branch and merge it with main branch.

```
$ git merge feature1
Already up to date.
```

Commands Used:

```
git branch new-feature
git checkout new-feature
echo "New feature" >> file.txt
git add .
git commit -m "Added new feature"
git checkout main
git merge new-feature
```

Advantages of Git

1. Distributed system (works offline)
2. Fast and efficient
3. Supports branching and merging
4. Easy collaboration
5. Tracks complete history

Challenges in Version Control

1. Learning curve for beginners
2. Merge conflicts
3. Requires proper management of branches